MUNJAZ | مُنجَز

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Project Proposal: Enhanced Campus Issue Reporting System (Munjaz/مُنجَز)

1. Introduction

Facility maintenance within university campuses is a critical aspect of ensuring a functional, safe, and efficient learning environment. Traditional reporting methods such as phone calls and emails often lead to miscommunication, delayed responses, and lack of transparency. To address these inefficiencies, the **Munjaz** project proposes a mobile application that leverages **Augmented Reality (AR)**, **Artificial Intelligence (AI)**, and **Internet of Things (IoT)** technologies to streamline issue reporting, predictive maintenance, and real-time tracking.

Recent studies highlight the growing relevance of AR and AI in maintenance contexts. AR provides interactive visual guidance that improves fault localization and user empowerment [1][6], while AI and predictive analytics enable proactive fault detection, reducing downtime and improving resource allocation [2]. Furthermore, IoT systems for smart campuses have proven essential in integrating data-driven insights for effective facility management [3][7].

1.1 Project Information

Field	Detail
Course Title	Mobile Computing (CS513)
Project Title	**Munjaz:** An AR and AI-Powered Mobile Solution for Smart Campus Facility Management
Topic Selection	IoT/Sensors for Monitoring
Rationale for Topic	Leveraging mobile sensors (Camera, Gyroscope) for AR localization and AI-based issue analysis.
Target Platform	Android
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Project Information 1 Table

2. Introduction and Problem Statement

2.1 The Current Problem

Current campus facility management systems face several challenges:

1. **Inaccurate Location Identification**: Users often fail to provide precise descriptions of fault locations, leading to wasted time and inefficient troubleshooting [1].
2. **Overload of Simple Issues**: Maintenance teams are frequently burdened with trivial problems that could be solved by the users themselves through guided instructions [6].
3. **Lack of Transparency**: Users have limited visibility into the status of their reports, creating frustration and dissatisfaction [3].
4. **Inefficient Resource Allocation**: Without smart task distribution, critical issues may be delayed due to poor prioritization [5].

These problems highlight the need for a digital, intelligent, and transparent facility management system.

2.2 Significance of the Problem

Applying advanced technologies like AR and AI is essential to shift facility maintenance from a reactive system to a proactive and efficient one. Our project addresses the core inefficiencies by automating the triage process and providing immediate support, ensuring issues are **Munjaz (مُنجَز)**, quickly and efficiently.



Figure 1 Munjaz App Interface for Issue Detection

3. Proposed Solution and Project Goal

3.1 The Solution: Munjaz (مُنجَز) Application

We propose developing a **server-client mobile application** called **Munjaz (مُنجَز)**. The system provides a unified platform that combines AR for spatial guidance and AI for intelligent issue analysis. **A campus user (student or staff member)** simply directs their phone camera toward the issue for immediate analysis.

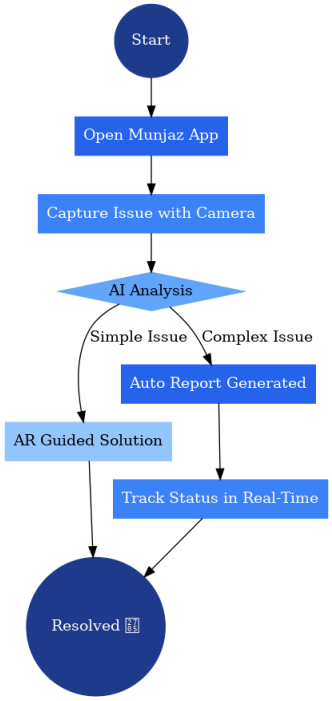


Figure 2 Munjaz Workflow Diagram

3.2 Core Objectives

- **Self-Service Empowerment (AR + AI):** Minor issues (e.g., a simple configuration) will be identified by AI and solved by overlaying AR-based step-by-step visual instructions on the user's device [4].
- **Accurate Reporting and Classification:** For complex problems, AI will classify the issue, attach AR-pinpointed location data and images, and forward the report to the correct department (maintenance, IT, or services) [1][2].
- **Real-Time Transparency:** The system will provide a server-client infrastructure where users can track issue progress from submission to resolution in real-time [3].

- **Efficient Resource Allocation:** Smart prioritization and predictive maintenance models will reduce unnecessary workloads and enhance the speed of response [2][5].
- **Enhanced Campus Experience:** By reducing the volume of trivial issues and accelerating repair times, Munjaz will contribute to an improved learning and working environment.

4. Technical Implementation and Mobile Computing Aspects

The project will satisfy the core requirements of the Mobile Computing course by focusing on the following technical components:

4.1 Mobile Platform and AR/AI Integration

The application will be developed for **Android**, leveraging **ARCore** for spatial tracking, surface detection, and real-world interaction. ARCore enables motion tracking and environmental understanding, crucial for anchoring visual guidance elements [1][4].
On the AI side, lightweight models (either on-device or server-based) will process captured images to classify issues and determine the repair approach. Recent frameworks show how AI-driven conversational systems such as ChatGPT can enhance AR-based maintenance workflows, providing contextual assistance to users [4].

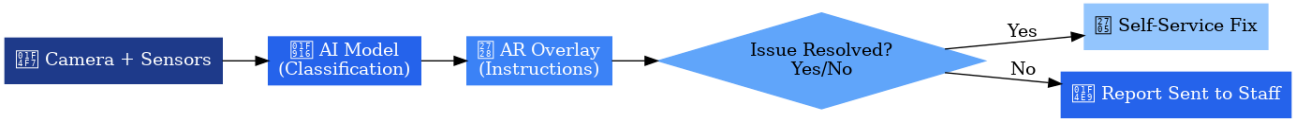


Figure 3 AR and AI Integration in Munjaz

The **AI component** (likely implemented on the server or via a lightweight mobile model) will handle image recognition and problem classification to determine if a self-service solution is applicable. This integration is crucial for the project's **"IoT/Sensors for Monitoring"** requirement, utilizing the camera and internal sensors.

4.2 Server-Client Environment

- **Client (Mobile App):** Handles user interaction, AR visualizations, AI pre-processing, and report submission.
- **Server (Backend):** Stores issue data in a scalable database (e.g., Firebase or MongoDB), manages AI models, tracks workflow status (New → Self-Service → In Progress → Resolved), and controls user roles.

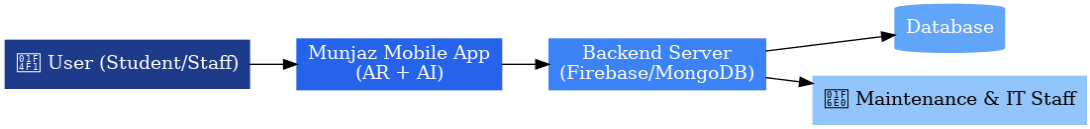


Figure 4 Client-Server Architecture

4.3 IoT and Predictive Maintenance

The system will integrate IoT-enabled sensors (e.g., temperature, vibration, energy usage) to support predictive maintenance strategies. This approach enables anomaly detection and preemptive interventions before failures occur [2][3][7].

4.4 Smart Decision-Making

By applying resource optimization frameworks [5], Munjaz will dynamically allocate tasks to available technicians, ensuring efficient use of resources and reducing resolution times.

5. Technical Value Added

The **integration of AR, AI, IoT, and predictive analytics** makes Munjaz a novel and practical solution for smart campus management. Unlike traditional reporting systems, Munjaz empowers users with **self-service capabilities**, ensures **precise issue classification**, and supports **proactive maintenance**. This convergence of technologies directly aligns with modern research in AR-based facility maintenance [1][6], predictive maintenance frameworks [2], and smart campus IoT applications [3][7].

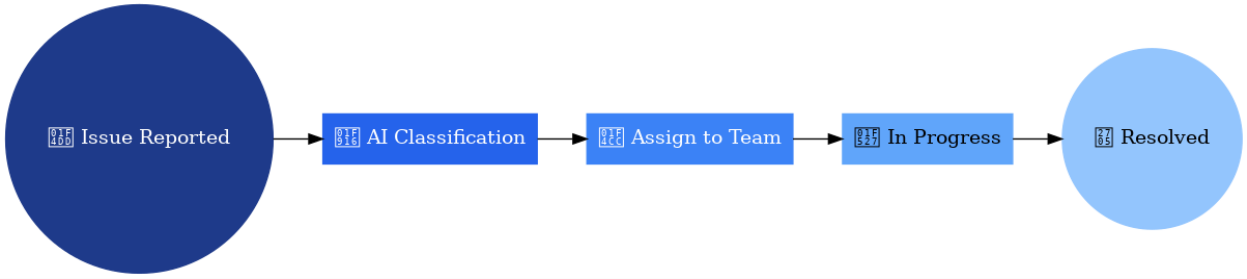


Figure 5 Smart Decision-Making Model

6. Conclusion

The **Munjaz (مُنجَز)** project presents a highly practical, technically innovative, and academically robust application for the Mobile Computing course. By integrating **Augmented Reality** for precise localization and **Artificial Intelligence** for intelligent problem-solving, this solution moves beyond traditional issue tracking to create a smart, self-service-enabled system. The proposed system effectively integrates the required **server-client architecture** and **mobile sensor monitoring** aspects. We are confident that this project is feasible within the given timeframe and will yield a distinguished application that directly enhances the campus environment.

7. References

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